Lecture #11: The Likelihood Function & Maximum-Likelihood Estimators

MATH 321 – Mathematical Statistics I

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The Likelihood Function

Suppose we collect data from a known family of distributions, but we don't know the parameters. How can we use this data to **estimate** the true value of the parameters?

For example, suppose I had been polling before the November 2025 gubernatorial election in Virginia between the Democrat Abigail Spanberger and the Republican Winsome Earle-Sears.

I can treat each voter's choice as a Bernoulli random variable, but how can I estimate the true proportion of voters who would vote for Spanberger?

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The Likelihood Function

Let $\boldsymbol{X} = [X_1, \ldots, X_N]^T$ be a random vector drawn from a joint PDF (or PMF) $f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\theta})$, and let $\boldsymbol{x} = [x_1, \ldots, x_N]^T$ be the realizations. The **likelihood function** is a function of the parameter $\boldsymbol{\theta}$ given the realizations $\boldsymbol{x}$:

$$\mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x}) \stackrel{\text{def}}{=} f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\theta}).$$

Given i.i.d. random variables $X_1, \ldots, X_N$ that all have the same PDF (or PMF) $f_X(x; \boldsymbol{\theta})$, the **likelihood function** is:

$$\mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x}) \stackrel{\text{def}}{=} \prod_{n=1}^{N} f_X(x_n; \boldsymbol{\theta}).$$

What are the constants in these equations? What are the variables?

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The Log-Likelihood Function

Given a set of i.i.d. random variables $X_1, \ldots, X_N$ with PDF (or PMF) $f_X(x; \boldsymbol{\theta})$, the **log-likelihood** is defined as

$$\log \mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x}) = \log f_{\boldsymbol{X}}(\boldsymbol{x}; \boldsymbol{\theta}) = \sum_{n=1}^{N} \log f_X(x_n; \boldsymbol{\theta}).$$

The **log-likelihood** is useful because it turns a product into a sum, which makes it much easier to differentiate. Since log is strictly increasing, $\mathcal{L}$ and $\log \mathcal{L}$ are maximized at the same value of $\boldsymbol{\theta}$.

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Example: The Log-Likelihood

Example: Find the log-likelihood of a sequence of i.i.d. Bernoulli random variables $X_1, \ldots, X_N$ with parameter θ.

Example: Find the log-likelihood of a sequence of i.i.d. Normal random variables $X_1, \ldots, X_N$ with mean μ and variance σ^2.

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The Maximum-Likelihood Estimator

Let $\mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x})$ be the likelihood function of the parameter $\boldsymbol{\theta}$ given the measurements $\boldsymbol{x} = [x_1, \ldots, x_N]^T$. The **maximum-likelihood estimate of the parameter $\boldsymbol{\theta}$** is a parameter value that maximizes the likelihood:

$$\widehat{\boldsymbol{\theta}}_{ML} \stackrel{\text{def}}{=} \arg\max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta} \mid \boldsymbol{x}).$$

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Example: Maximum-Likelihood Estimators

Example: Find the ML estimate for a set of i.i.d. Bernoulli random variables $\{X_1, \ldots, X_N\}$ with $X_n \sim \text{Bernoulli}(\theta)$ for $n = 1, \ldots, N$.

Example: Find the ML estimates for a set of i.i.d. Normal random variables $\{X_1, \ldots, X_N\}$ with $X_n \sim \text{Normal}(\mu, \sigma^2)$ for $n = 1, \ldots, N$.

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Estimate vs. Estimator

An **estimate** is a **number**, e.g. $\widehat{\theta}_{ML} = \dfrac{1}{N}\sum_{n=1}^{N} x_n$. It is a realization of an estimator.

An **estimator** is a **random variable**, e.g. $\widehat{\Theta}_{ML} = \dfrac{1}{N}\sum_{n=1}^{N} X_n$. It takes a set of random variables and generates another random variable.

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Unbiased Estimators

An estimator $\widehat{\Theta}$ of θ is **unbiased** if

$$E[\widehat{\Theta}] = \theta.$$

Example: Let $X_1, \ldots, X_N$ be i.i.d. Normal random variables with unknown mean μ and unknown variance σ^2. We have shown that the ML estimators are:

$$\widehat{\mu}_{ML} = \frac{1}{N}\sum_{n=1}^{N} X_n \qquad \text{and} \qquad \widehat{\sigma}^2_{ML} = \frac{1}{N}\sum_{n=1}^{N} \left(X_n - \widehat{\mu}_{ML}\right)^2.$$

It is easy to show that $E[\widehat{\mu}_{ML}] = \mu$. How about $\widehat{\sigma}^2_{ML}$? Is it an unbiased estimator?

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Consistent Estimators

A sequence of estimators $\widehat{\Theta}_N$ of θ is **consistent** if $\widehat{\Theta}_N \xrightarrow{p} \theta$, i.e., for every $\varepsilon > 0$,

$$\lim_{N\to\infty} P\left[\left|\widehat{\Theta}_N - \theta\right| > \varepsilon\right] = 0.$$

Example: If $V(X) < \infty$, the sample mean $\overline{X}_N = \dfrac{1}{N}\displaystyle\sum_{n=1}^{N} X_n$ is a consistent estimator of $E(X)$. Why?

Example: If $E(X^4) < \infty$, the sample variance $\widehat{\sigma}^2_{ML} = \dfrac{1}{N}\displaystyle\sum_{n=1}^{N}\left(X_n - \overline{X}_N\right)^2$ is a consistent estimator of $V(X)$. Why?

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